

3 Nitrogen dioxide, NO_2 , is an atmospheric pollutant and is formed in car engines.

(a) Explain how nitrogen dioxide is formed in car engines.

.....
 [2]

(b) Nitrogen dioxide causes respiratory problems.

State one **other** adverse effect of nitrogen dioxide.

..... [1]

(c) Nitrogen dioxide emissions can be reduced by adding an aqueous solution of urea, $(\text{NH}_2)_2\text{CO}$, to car exhaust gases.

The heat of the exhaust gases breaks down the urea into simpler substances.

(i) Name the type of reaction which occurs when a substance is heated and breaks down into simpler substances.

..... [1]

(ii) One molecule of urea breaks down to form one molecule of ammonia and one other molecule.

Complete the chemical equation to show the formula of the other molecule formed in this reaction.



(iii) State the test for ammonia.

test

observations

[2]

(d) The ammonia formed reacts with nitrogen dioxide to form nitrogen and water.

(i) Balance the equation for this reaction.



(ii) State how the equation shows that the nitrogen in nitrogen dioxide is reduced.

..... [1]

- (iii) This reaction is a redox reaction.

State the meaning of the term *redox*.

..... [1]

- (e) 135 moles of urea, $(\text{NH}_2)_2\text{CO}$, is stored in the tank of a car.

Calculate the mass, in kg, of the stored $(\text{NH}_2)_2\text{CO}$.

mass of $(\text{NH}_2)_2\text{CO}$ = kg
[2]

- (f) Another oxide of nitrogen formed in car engines is nitrogen monoxide, NO. A catalytic converter removes NO by reacting it with a gas formed by incomplete combustion of the fuel. Two non-toxic gases are formed.

- (i) Name the gas formed by incomplete combustion of the fuel.

..... [1]

- (ii) Name the **two** non-toxic gases formed.

..... and [1]

[Total: 15]

1 A list of substances is shown.

aluminium oxide	carbon dioxide	chlorine	diamond	ethanol
glucose	iron(III) oxide	limestone	nitrogen	oxygen

Answer the questions using the list of substances.

Each substance may be used once, more than once or not at all.

State which of the substances:

(a) is a reactant in photosynthesis

..... [1]

(b) is the main constituent of bauxite

..... [1]

(c) are **two** products of fermentation

..... and [2]

(d) is used as a fuel

..... [1]

(e) is a gas used to convert iron into steel

..... [1]

(f) is a greenhouse gas

..... [1]

(g) is a gas that is approximately 78% of clean, dry air

..... [1]

(h) is a form of carbon.

..... [1]

[Total: 9]

1 A list of substances is shown.

aluminium oxide	carbon dioxide	carbon monoxide	chlorine	copper
glucose	iron(III) oxide	limestone	nitrogen	oxygen

Answer the questions using the substances in the list.

Each substance may be used once, more than once or not at all.

State which substance is:

(a) a product of respiration

..... [1]

(b) the main constituent of hematite

..... [1]

(c) an element which has a sulfate that is used to test for water

..... [1]

(d) a colourless toxic gas

..... [1]

(e) a reactant in fermentation

..... [1]

(f) a reducing agent in the extraction of iron

..... [1]

(g) a conductor of electricity when solid

..... [1]

(h) a gas that is approximately 21% of clean, dry air.

..... [1]

[Total: 8]

2 The elements shown are gases at room temperature and pressure.

hydrogen
nitrogen
oxygen
chlorine

(a) State which **one** of these gases is green.

..... [1]

(b) The gases shown exist as diatomic molecules.

State the name of **another** element which has diatomic molecules and is a gas at room temperature and pressure.

..... [1]

(c) When separate samples of each of these gases are placed in a container they will diffuse.

(i) Describe why these gases diffuse.

..... [1]

(ii) State which of these four gases has the highest rate of diffusion.

Explain your answer.

gas

explanation

..... [2]

(d) Nitrogen, oxygen and other substances are found in clean, dry air.

(i) State the percentage of nitrogen in clean, dry air.

..... [1]

(ii) Other than nitrogen and oxygen, identify another element found in clean, dry air.

..... [1]

(iii) Identify a compound found in clean, dry air.

..... [1]

(iv) Nitrogen and oxygen can be separated from liquid air.

State the name of this process.

..... [2]

[Total: 10]

3 (a) Sulfuric acid is made from sulfur in a four-stage process.

stage 1 Sulfur is converted into sulfur dioxide.

stage 2 Sulfur dioxide is converted into sulfur trioxide.

stage 3 Sulfur trioxide is converted into oleum.

stage 4 Oleum is converted into sulfuric acid.

(i) How is sulfur converted into sulfur dioxide in **stage 1**?

..... [1]

(ii) Describe how sulfur dioxide is converted into sulfur trioxide in **stage 2**.

Your answer should include:

- an equation for the reaction
- the temperature used
- the name of the catalyst used.

.....

 [3]

(iii) The reaction in **stage 2** can reach equilibrium.

What is meant by the term *equilibrium*?

.....

 [2]

(b) Sulfur trioxide is converted into oleum, $\text{H}_2\text{S}_2\text{O}_7$, in **stage 3**.

What is sulfur trioxide reacted with to convert it into oleum?

..... [1]

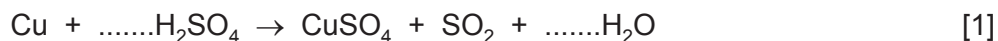
(c) Oleum is converted into sulfuric acid in **stage 4**.

Write a chemical equation for the conversion of oleum, $\text{H}_2\text{S}_2\text{O}_7$, into sulfuric acid.

..... [2]

- (d) When copper is reacted with hot concentrated sulfuric acid, sulfur dioxide gas is formed.

Balance the chemical equation for this reaction.



- (e) Sulfur dioxide is a reducing agent.

Give the colour change that occurs when excess sulfur dioxide is bubbled into acidified aqueous potassium manganate(VII).

starting colour of the solution

final colour of the solution [1]

- (f) When sulfuric acid reacts with ammonia the salt produced is ammonium sulfate.

Write the chemical equation for this reaction.

..... [2]

- (g) Barium sulfate is an insoluble salt.

Barium sulfate can be made from aqueous ammonium sulfate using a precipitation reaction.

- (i) Name a solution that can be added to aqueous ammonium sulfate to produce a precipitate of barium sulfate.

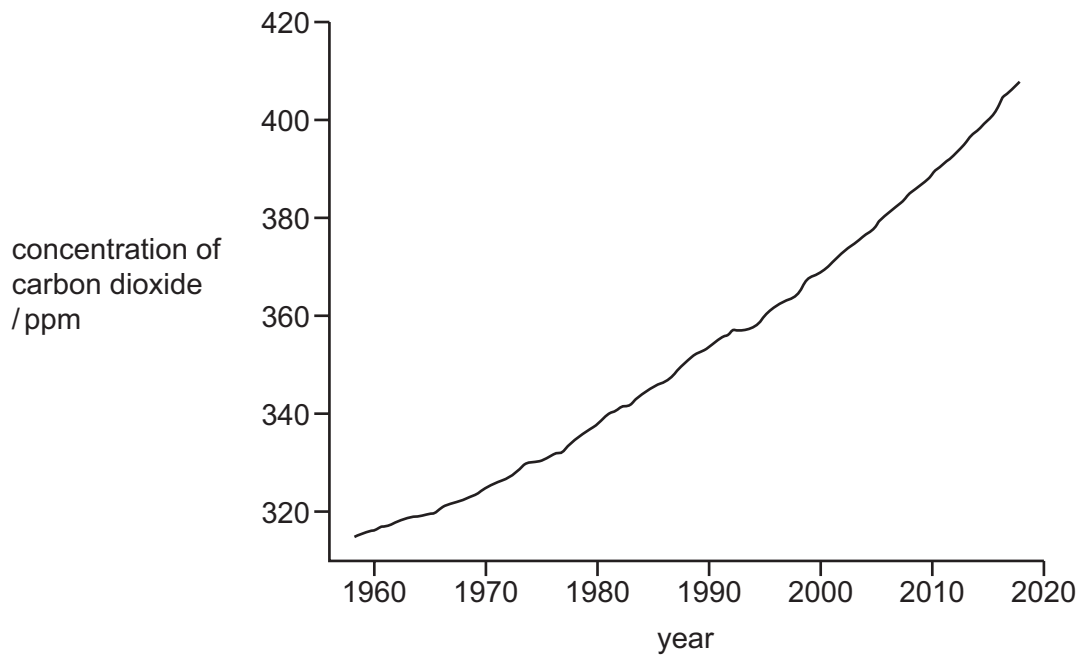
..... [1]

- (ii) Write an ionic equation for this precipitation reaction. Include state symbols.

..... [2]

[Total: 16]

- (d) The graph shows the concentration of carbon dioxide in the atmosphere over a 60-year period, measured in parts per million (ppm).



The data shown in the graph is of global concern.

Explain why.

.....

.....

.....

.....

.....

..... [3]

- (e) Name the process in the carbon cycle by which plants remove carbon dioxide from the atmosphere.

..... [1]

[Total: 10]

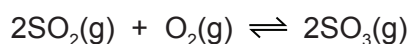
- (b) Sulfuric acid is manufactured using the Contact process. This manufacture involves four stages.

- (i) **Stage 1** involves the combustion of sulfur to form sulfur dioxide.

Write the chemical equation for **stage 1**.

..... [1]

- (ii) The equation for **stage 2** is shown.



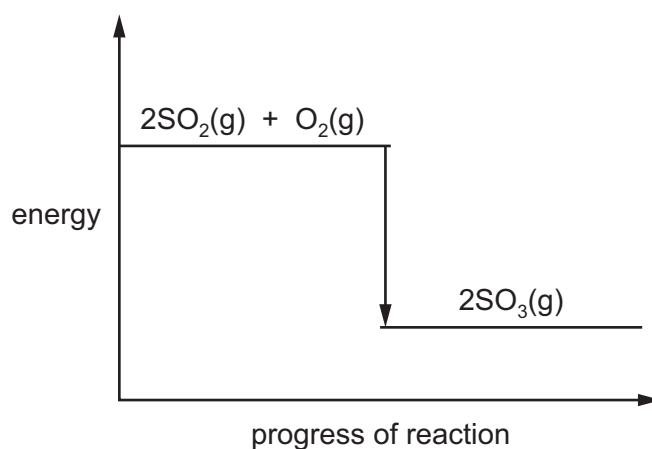
The reaction can reach equilibrium.

Explain what is meant by the term *equilibrium*.

.....

 [2]

- (iii) The energy level diagram for the forward reaction in **stage 2** is shown.



Explain what the diagram shows about the energy changes in the forward reaction.

.....

 [2]

(c) In **stage 3** sulfur trioxide, SO_3 , is converted to oleum, $\text{H}_2\text{S}_2\text{O}_7$.

In **stage 4** oleum reacts to form sulfuric acid, H_2SO_4 .

State what oleum reacts with in **stage 4**.

..... [1]

(d) A sample of sulfuric acid, H_2SO_4 , has a concentration of 0.75 mol/dm^3 .

Calculate the concentration of sulfuric acid in g/dm^3 .

..... g/dm^3 [2]

[Total: 15]

1 The names of eight substances are given.

aluminium oxide

calcium oxide

ethanol

nitrogen

iron(III) oxide

methane

oxygen

silicon(IV) oxide

Answer the following questions about these substances.

Each substance may be used once, more than once or not at all.

State which substance is:

(a) the main constituent of natural gas

..... [1]

(b) a reactant in respiration

..... [1]

(c) the main constituent of bauxite

..... [1]

(d) a product of photosynthesis

..... [1]

(e) a greenhouse gas

..... [1]

(f) a macromolecular solid.

..... [1]

[Total: 6]

(c) The manufacture of sulfuric acid by the Contact process occurs in four stages.

stage 1 Molten sulfur is burned in air to produce sulfur dioxide gas.

stage 2 Sulfur dioxide is reacted with oxygen to form sulfur trioxide.

stage 3 Sulfur trioxide is combined with concentrated sulfuric acid to form oleum, $\text{H}_2\text{S}_2\text{O}_7$.

stage 4 Oleum is added to water to form sulfuric acid.

(i) Complete the chemical equation for **stage 1** by adding the appropriate state symbols.



(ii) Name the catalyst used in **stage 2** and state the temperature used.

catalyst

temperature °C

[2]

(iii) Write chemical equations for the reactions in **stage 3** and **stage 4**.

stage 3

stage 4

[2]

(d) Sulfur dioxide is a toxic gas.

(i) State one **environmental** reason why sulfur dioxide should **not** be released into the atmosphere.

..... [1]

(ii) Describe the test for sulfur dioxide.

test

.....

observations

.....

[2]

- (e) Sulfur dioxide reacts with aqueous sodium sulfite to produce a compound with the following composition by mass: 29.1% Na, 40.5% S and 30.4% O.

Calculate the empirical formula of this compound.

empirical formula = [3]

[Total: 16]